Uniqueness of Yangians and Twisted Yangians

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The uniqueness problem

The question

How much of a quantum algebra is determined by its classical limit and its tensor-product structure?

1. Current algebras and homogeneous quantization.
2. Drinfeld's original J-relations and the GWX uniqueness theorem.
3. Our work on coideal quantizations for symmetric pairs.

Lie bialgebra

A Lie bialgebra is a Lie algebra L together with a linear map

$$\delta : L \longrightarrow L \otimes L \qquad \text{(the Lie cobracket)}$$

satisfying three conditions:

1. **Skew-symmetry:** $\delta^{21}(x) = -\delta(x)$.
2. **Co-Jacobi:** the dual operation is a Lie bracket:

$$\langle [\xi, \eta]_*, x \rangle = \langle \xi \otimes \eta, \delta(x) \rangle, \qquad \xi, \eta \in L^*.$$

3. **Compatibility with the original bracket:**

$$\delta([x, y]) = x \cdot \delta(y) - y \cdot \delta(x),$$

where $x \cdot (a \otimes b) = [x, a] \otimes b + a \otimes [x, b]$.

A Lie algebra and a compatible Lie bracket on its dual.

Current Lie bialgebra

Inside the loop algebra $\mathfrak{g}[u, u^{-1}]$, consider

$$L_+ = \mathfrak{g}[u], \qquad L_- = u^{-1}\mathfrak{g}[u^{-1}].$$

An invariant form $(\ ,\ )$ on $\mathfrak{g}$ pairs positive and negative currents:

$$\langle xu^r, yu^s \rangle = -(x, y)\,\mathrm{Res}_{u=0}(u^{r+s}\,du) = -(x, y)\delta_{r+s,-1}.$$

Transpose the bracket on negative currents

$$\langle \delta(P), A \otimes B \rangle = \langle P, [A, B] \rangle, \qquad P \in L_+, \quad A, B \in L_-.$$

Since $[au^{-1}, bu^{-1}] = [a, b]u^{-2}$, we obtain

$$\langle \delta(xu), au^{-1} \otimes bu^{-1} \rangle = -(x, [a, b]).$$

Current Lie bialgebra

For dual bases $\{x_i\}, \{x^i\}$, set $\Omega = \sum_i x_i \otimes x^i$. Invariance of the form gives

$$\big([x \otimes 1, \Omega], a \otimes b\big)_{\otimes} = ([x, b], a) = -(x, [a, b]).$$

Comparing with the previous slide yields $\delta(xu) = [x \otimes 1, \Omega]$. More generally, identifying the two current variables with u, v,

$$\delta(xu^r) = [x \otimes 1, \Omega]\, \frac{u^r - v^r}{u - v} \quad (r \geqslant 1), \qquad \delta(x) = 0.$$

The quotient is the polynomial $u^{r-1} + u^{r-2}v + \cdots + v^{r-1}$.

Example

$(e, f) = 1$ and $(h, h) = 2$ in $\mathfrak{sl}_2$

$$\Omega = e \otimes f + f \otimes e + \tfrac{1}{2} h \otimes h,$$
$$\delta(eu) = [e, f] \otimes e + \tfrac{1}{2}[e, h] \otimes h = h \otimes e - e \otimes h.$$

Homogeneous quantization

We seek a graded Hopf algebra H over $\mathbb{C}[\hbar]$ with

$$H/\hbar H \cong \mathrm{U}(\mathfrak{g}[u]), \qquad \deg \hbar = 1.$$

- Freeness over $\mathbb{C}[\hbar]$ preserves the classical basis.
- Multiplication and coproduct respect the grading.
- Their first-order coproduct asymmetry recovers a fixed cobracket:

$$\delta(\bar{z}) = \frac{\Delta(z) - \Delta^{\mathrm{op}}(z)}{\hbar} \bmod \hbar, \qquad \bar{z} \in \mathfrak{g}[u].$$

Here Δ^{op} interchanges the tensor factors.

Drinfeld's original J-presentation

The Yangian $\mathrm{Y}_\hbar(\mathfrak{g})$ is the unital associative algebra generated by elements x and $J(x)$ for $x \in \mathfrak{g}$ with the defining relations:

$$
\begin{aligned}
&xy - yx = [x, y] \text{ for all } x, y \in \mathfrak{g},\ J \text{ is linear in } x \in \mathfrak{g}, \quad J([x, y]) = [x, J(y)],\\
&\big[J(x), J([y, z])\big] + \big[J(z), J([x, y])\big] + \big[J(y), J([z, x])\big]\\
&\qquad\qquad = \hbar^2 \sum_{a,b,c \in \Lambda} \big([x, \xi_a], \big[[y, \xi_b], [z, \xi_c]\big]\big)\big\{\xi_a, \xi_b, \xi_c\big\},\\
&\big[[J(x), J(y)], [z, J(w)]\big] + \big[[J(z), J(w)], [x, J(y)]\big]\\
&= \hbar^2 \sum_{a,b,c \in \Lambda} \Big(\big([x, \xi_a], \big[[y, \xi_b], [[z, w], \xi_c]\big]\big)\\
&\qquad\qquad + \big([z, \xi_a], \big[[w, \xi_b], [[x, y], \xi_c]\big]\big)\Big)\big\{\xi_a, \xi_b, J(\xi_c)\big\},
\end{aligned}
$$

where $\{\xi_a\}_{a \in \Lambda}$ is an orthonormal basis of $\mathfrak{g}$, Λ being a fixed indexing set of size $\dim \mathfrak{g}$, and $\{A_1, A_2, A_3\} = \frac{1}{24} \sum_{\pi \in \mathfrak{S}_3} A_{\pi(1)} A_{\pi(2)} A_{\pi(3)}$.

How to find the relations

The quantum tensor-product rule

$\Delta(x) = x \otimes 1 + 1 \otimes x$
$\Delta(J(x)) = J(x) \otimes 1 + 1 \otimes J(x) + \frac{\hbar}{2}[x \otimes 1, \Omega]$.

$\triangle$ is an algebra homomorphism

The coproduct $\Delta(J(x)) = J(x) \otimes 1 + 1 \otimes J(x) + \frac{\hbar}{2}[x \otimes 1, \Omega]$ should define an algebra homomorphism.

The uniqueness theorem of GWX

Gautam–Wendlandt–Xu (2025)

The rational Lie bialgebra $\mathfrak{g}[u]$, for complex simple $\mathfrak{g}$, admits at most one homogeneous Hopf quantization, up to isomorphism.

- Start with any homogeneous quantization H.
- Normalize lifts $J(x)$ of xu using cohomological methods.
- Force a reduced family of relations using the coproduct.

Together with known existence and PBW, this identifies H with the Yangian.

S. Gautam, C. Wendlandt, S. Xu, *On the uniqueness of Yangians*, arXiv:2506.21904.

Uniqueness

Theorem [GWX]

Let H be a homogeneous quantization of $(\mathfrak{g}[u], \delta)$. Then there exists a $\mathfrak{g}$-module homomorphism $J : \mathfrak{g} \to H_1$ satisfying $J(x) = xu$ modulo $\hbar$ and

$$\Delta(J(x)) = J(x) \otimes 1 + 1 \otimes J(x) + \frac{\hbar}{2}[x \otimes 1, \Omega]$$

for each $x \in \mathfrak{g}$. Moreover, if $J' : \mathfrak{g} \to H_1$ is another such map, then there is $\lambda \in \mathbb{C}$ such that $J(x) - J'(x) = \hbar\lambda x$ for all $x \in \mathfrak{g}$.

- Take any $f : \mathfrak{g} \to H_1$ such that $f(x) = xu \mod \hbar H_0$ for each $x \in \mathfrak{g}$
- There exists a linear map $\varphi : \mathfrak{g} \to U(\mathfrak{g})$ such that $J(x) := f(x) + \hbar\varphi(x)$ satisfies the properties
- Use Whitehead's first lemma and Cartier's computation of the cohomology of the coalgebra complex associated to $\mathrm{U}(\mathfrak{g})$

The reduced Cartan relation

Fix a Cartan subalgebra $\mathfrak{h}$ and dual positive and negative root vectors. Put

$$\nu(t) = \frac{1}{2} \sum_{\alpha > 0} \alpha(t) x_\alpha^- x_\alpha^+ \in \mathrm{U}(\mathfrak{g}), \qquad t \in \mathfrak{h}.$$

A complete reduced presentation, for $\mathfrak{g} \ncong \mathfrak{sl}_2$

Besides the Lie and module relations, it is enough to impose

$$[\mathcal{J}(t), \mathcal{J}(t')] = \hbar^2 [\nu(t'), \nu(t)] \qquad (t, t' \in \mathfrak{h}).$$

For $\mathfrak{sl}_2$, one needs

$$[[\mathcal{J}(e), \mathcal{J}(f)], \mathcal{J}(h)] = \hbar^2 (f\mathcal{J}(e) - \mathcal{J}(f)e)h \quad \text{if} \quad \mathfrak{g} \cong \mathfrak{sl}_2$$

It is equivalent to the minimalistic presentation of [Guay-Nakajima-Wendlandt].

Key lemma

Primitive element

Let H be a homogeneous quantization of $(\mathfrak{g}[u], \delta)$. Then the Lie algebra $\mathrm{Prim}(H)$ of primitive elements in H is equal to $\mathfrak{g} \otimes_{\mathbb{C}} \mathbb{C}[\hbar]$:

$$\mathrm{Prim}(H) := \{y \in H : \Delta(y) = y \otimes 1 + 1 \otimes y\} = \mathfrak{g} \otimes_{\mathbb{C}} \mathbb{C}[\hbar].$$

Symmetric pairs

Let $\omega \neq \mathrm{id}$ be an involution of the complex simple Lie algebra $\mathfrak{g}$. Write

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}, \qquad \omega|_{\mathfrak{k}} = \mathrm{id}, \qquad \omega|_{\mathfrak{p}} = -\mathrm{id}.$$

Then

$$[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}, \qquad [\mathfrak{k}, \mathfrak{p}] = \mathfrak{p}, \qquad [\mathfrak{p}, \mathfrak{p}] = \mathfrak{k}.$$

Example: $\mathfrak{sl}_2$

Take $\omega(h) = h, \omega(e) = -e, \omega(f) = -f$. Thus

$$\mathfrak{k} = \mathbb{C}h, \qquad \mathfrak{p} = \mathbb{C}e \oplus \mathbb{C}f.$$

Twisted current algebra

Extend ω to $\mathfrak{g}[u]$ by

$$\check{\omega}(xu^r) = (-1)^r \omega(x) u^r.$$

Its fixed Lie algebra is

$$\mathfrak{L} = \mathfrak{g}[u]^{\check{\omega}} = \mathfrak{k}[u^2] \oplus u\mathfrak{p}[u^2].$$

Example: $\mathfrak{sl}_2$

degree	0	1	2	3
$\mathfrak{g}[u]^{\check{\omega}}$	h	eu, fu	hu^2	eu^3, fu^3

For instance, $[eu, fu] = hu^2$ and $[hu^2, eu] = 2eu^3$.

In general, $\mathfrak{g}[u]^{\check{\omega}}$ is generated as a Lie algebra by $\mathfrak{k}$ and $u\mathfrak{p}$.

Coideal subalgebras

Definition

A subalgebra $B \subset \mathrm{Y}_\hbar(\mathfrak{g})$ is a right coideal if

$$\Delta(B) \subset B \otimes_{\mathbb{C}[\hbar]} \mathrm{Y}_\hbar(\mathfrak{g}).$$

If M is a B-module and V is a Yangian module, the coproduct makes $M \otimes V$ a B-module.

The classical $\mathfrak{sl}_2$ example

$$\delta(eu) = h \otimes e - e \otimes h.$$

Here $h \in \mathfrak{g}[u]^{\breve{\omega}}$, whereas $e \notin \mathfrak{g}[u]^{\breve{\omega}}$.
Thus the cobracket takes values in $\mathfrak{g}[u]^{\breve{\omega}} \otimes \mathfrak{g}[u] + \mathfrak{g}[u] \otimes \mathfrak{g}[u]^{\breve{\omega}}$.

Homogeneous coideal quantization

Fix the Yangian $\mathrm{Y}_\hbar(\mathfrak{g})$ and the natural inclusion

$$\mathrm{U}(\mathfrak{g}[u]^{\check{\omega}}) \subset \mathrm{U}(\mathfrak{g}[u]).$$

Definition [L-Wang-Weekes]

A homogeneous coideal quantization is a graded right coideal subalgebra $B \subset \mathrm{Y}_\hbar(\mathfrak{g})$, free over $\mathbb{C}[\hbar]$, such that the natural reduction identifies

$$B/\hbar B \cong \mathrm{U}(\mathfrak{g}[u]^{\check{\omega}}) \subset H/\hbar H \cong \mathrm{U}(\mathfrak{g}[u]).$$

We ask whether these conditions determine B.

Previous results: [Belliard-Regelskis14]

Definition 5.3. Let $(\tilde{\mathfrak{g}}, \delta)$ be a Lie bi-algebra and $\left(\tilde{\mathfrak{g}}^\theta, \tau\right)$ be a left Lie coideal. We say that a left coideal subalgebra $\left(\mathcal{U}_\hbar\left(\tilde{\mathfrak{g}}^\theta\right), \Delta_\hbar\right)$ is a quantization of $\left(\tilde{\mathfrak{g}}^\theta, \tau\right)$, or that $\left(\tilde{\mathfrak{g}}^\theta, \tau\right)$ is the quasi-classical limit of $\left(\mathcal{U}_\hbar\left(\tilde{\mathfrak{g}}^\theta\right), \Delta_\hbar\right)$, if it is a topologically free $\mathbb{C}[[\hbar]]$ module and:

1) $(\mathcal{U}_\hbar(\tilde{\mathfrak{g}}), \Delta_\hbar)$ is a quantization of $(\tilde{\mathfrak{g}}, \delta)$;
2) $\mathcal{U}_\hbar\left(\tilde{\mathfrak{g}}^\theta\right)/\hbar\mathcal{U}_\hbar\left(\tilde{\mathfrak{g}}^\theta\right)$ is isomorphic to $\mathcal{U}\left(\tilde{\mathfrak{g}}^\theta\right)$ as a Lie algebra;
3) $\left(\mathcal{U}_\hbar\left(\tilde{\mathfrak{g}}^\theta\right), \Delta_\hbar\right)$ is a left coideal of $(\mathcal{U}_\hbar(\tilde{\mathfrak{g}}), \Delta_\hbar)$;
4) for any $x \in \tilde{\mathfrak{g}}^\theta$ and any $X \in \mathcal{U}_\hbar\left(\tilde{\mathfrak{g}}^\theta\right)$ equal to $x \pmod{\hbar}$ one has

$$\left(\Delta_\hbar(X) - (\varphi(X) \otimes 1 + 1 \otimes X)\right)/\hbar \sim \tau(x) \pmod{\hbar}$$

with φ the natural embedding $\mathcal{U}_\hbar\left(\tilde{\mathfrak{g}}^\theta\right) \hookrightarrow \mathcal{U}_\hbar(\tilde{\mathfrak{g}})$.

Previous results: [Belliard-Regelskis14]

eigenspace of θ. Let $\{X_\alpha, Y_p\}$ be a symmetric space basis of $\mathfrak{g}$ with respect to θ. There is, up to isomorphism, a unique homogeneous quantization $\mathcal{Y}\big(\mathfrak{g},\mathfrak{g}^\theta\big)^{\mathrm{tw}} := \mathcal{U}_\hbar\big(\mathfrak{g}[u]^\theta\big)$ of $\big(\mathfrak{g}[u]^\theta, \tau\big)$, where τ is given by (4.5). It is topologically generated by elements X_α, $\mathcal{B}(Y_p)$ with the defining relations:

$$[X_\alpha, X_\beta] = f_{\alpha\beta}{}^{\gamma} X_\gamma, \qquad [X_\alpha, \mathcal{B}(Y_p)] = g_{\alpha p}{}^{q} \mathcal{B}(Y_q),$$

$$\mathcal{B}(aY_p + bY_q) = a\mathcal{B}(Y_p) + b\mathcal{B}(Y_q), \tag{5.10}$$

$$[\mathcal{B}(Y_p), \mathcal{B}(Y_q)] + \sum_\alpha (\bar{\mathfrak{c}}_{(\alpha)})^{-1} w_{pq}{}^{\alpha} w_\alpha{}^{rs} [\mathcal{B}(Y_r), \mathcal{B}(Y_s)] = \hbar^2 \sum_{\lambda,\mu,\nu} \Lambda_{pq}^{\lambda\mu\nu} \{X_\lambda, X_\mu, X_\nu\}, \tag{5.11}$$

$$\begin{aligned}[[\mathcal{B}(Y_p), \mathcal{B}(Y_q)], \mathcal{B}(Y_r)] + 2\mathfrak{c}_\mathfrak{g}^{-1} \sum_\alpha (\kappa_\mathfrak{m})^{tu} w_{pq}{}^{\alpha} g_{r\alpha}{}^{s} [[\mathcal{B}(Y_s), \mathcal{B}(Y_t)], \mathcal{B}(Y_u)] \\ = \hbar^2 \sum_{\lambda,\mu,u} \Upsilon_{pqr}^{\lambda\mu u} \{X_\lambda, X_\mu, \mathcal{B}(Y_u)\},\end{aligned} \tag{5.12}$$

for all $X_\alpha, Y_p \in \mathfrak{g}$ and $a, b \in \mathbb{C}$. Here

$$\Lambda_{pq}^{\lambda\mu\nu} = \tfrac{1}{3}\Big(g^{\mu t}{}_p g^{\lambda u}{}_q + \sum_\alpha (\bar{\mathfrak{c}}_{(\alpha)})^{-1} w_{pq}{}^{\alpha} w_\alpha{}^{rs} g^{\mu t}{}_r g^{\lambda u}{}_s \Big) w_{tu}{}^{\nu}, \tag{5.13}$$

$$\begin{aligned}\Upsilon_{pqr}^{\lambda\mu u} = \tfrac{1}{4} \sum_\alpha \Big(w_{st}{}^{\alpha} g_p{}^{\lambda s} g_q{}^{\mu t} g_{\alpha r}{}^{u} + \sum_\beta w_{pq}{}^{\alpha} f_\alpha{}^{\lambda\beta} g_r{}^{\mu s} g_{\beta s}{}^{u} \Big) \\ + \tfrac{1}{2} \mathfrak{c}_\mathfrak{g}^{-1} \sum_{\alpha,\gamma} (\kappa_\mathfrak{m})^{vx} w_{pq}{}^{\gamma} g_{r\gamma}{}^{y} \Big(w_{st}{}^{\alpha} g_y{}^{\lambda s} g_v{}^{\mu t} g_{\alpha x}{}^{u} + \sum_\beta w_{yv}{}^{\alpha} f_\alpha{}^{\lambda\beta} g_x{}^{\mu s} g_{\beta s}{}^{u} \Big).\end{aligned} \tag{5.14}$$

Existence and uniqueness

Our theorem

Let $\omega \neq \mathrm{id}$ be an involution of a complex simple Lie algebra $\mathfrak{g}$.

1. Homogeneous coideal quantizations of $\mathrm{U}(\mathfrak{g}[u]^{\tilde{\omega}})$ exist.
2. Every such subalgebra of $\mathrm{Y}_\hbar(\mathfrak{g})$ is a unique $B_{a,K}$, with
$$(a, K) \in \mathbb{C} \times \mathfrak{z}(\mathfrak{k}).$$
3. All $B_{a,K}$ are isomorphic as graded $\mathbb{C}[\hbar]$-algebras.

The theorem includes $\mathfrak{sl}_2$ and all nontrivial symmetric pairs.

Generators and coproduct

Put $C_{\mathfrak{k}} = \sum_r X_r X^r$ and $\Omega_{\mathfrak{k}} = \sum_r X_r \otimes X^r$ for dual bases of $\mathfrak{k}$.

The lift of yu, for $y \in \mathfrak{p}$

$$b_{a,K}(y) = J(y) - \frac{\hbar}{4}[y, C_{\mathfrak{k}}] + \hbar\big(ay + [K, y]\big).$$

The algebra $B_{a,K}$ is generated by $\mathfrak{k}$ and $b_{a,K}(\mathfrak{p})$. For $x \in \mathfrak{k}$ and $y \in \mathfrak{p}$,

$$[x, b_{a,K}(y)] = b_{a,K}([x, y]),$$
$$\Delta(b_{a,K}(y)) = b_{a,K}(y) \otimes 1 + 1 \otimes b_{a,K}(y) - \hbar[1 \otimes y, \Omega_{\mathfrak{k}}].$$

Correspondence with Drinfeld presentation (split case):

$$h_{i,1} \mapsto 2b_{0,0}(\xi_i) + \tfrac{1}{2} \sum_{\alpha \in \Phi^+} (\alpha, \alpha_i)(b_\alpha)^2.$$

Example: the twisted Yangian for $\mathfrak{sl}_2$

Take $\mathfrak{k} = \mathbb{C}h$, $\mathfrak{p} = \mathbb{C}e \oplus \mathbb{C}f$, and $a = K = 0$. Since $(h, h) = 2$, we have $C_{\mathfrak{k}} = \frac{1}{2}h^2$. Hence

$$b(e) = \mathcal{J}(e) + \frac{\hbar}{4}(eh + he), \qquad b(f) = \mathcal{J}(f) - \frac{\hbar}{4}(fh + hf).$$

The coideal coproduct

$$\begin{aligned}
\Delta(h) &= h \otimes 1 + 1 \otimes h, \\
\Delta(b(e)) &= b(e) \otimes 1 + 1 \otimes b(e) + \hbar h \otimes e, \\
\Delta(b(f)) &= b(f) \otimes 1 + 1 \otimes b(f) - \hbar h \otimes f.
\end{aligned}$$

Every first tensor factor belongs to $B = \mathbb{C}[\hbar]\langle h, b(e), b(f)\rangle$.

The correction in the $\mathfrak{sl}_2$ example

The Yangian coproduct gives

$$\Delta(\mathcal{J}(e)) = \mathcal{J}(e) \otimes 1 + 1 \otimes \mathcal{J}(e) + \frac{\hbar}{2}(h \otimes e - e \otimes h).$$

The first tensor factor e does not belong to the degree-zero algebra $\mathrm{U}(\mathfrak{k}) = \mathbb{C}[h]$. Using the primitive coproducts of e, h,

$$\Delta(eh + he) = (eh + he) \otimes 1 + 1 \otimes (eh + he) + 2(e \otimes h + h \otimes e).$$

Adding $\frac{\hbar}{4}(eh + he)$

$$\frac{\hbar}{2}(h \otimes e - e \otimes h) + \frac{\hbar}{2}(e \otimes h + h \otimes e) = \hbar h \otimes e.$$

Key lemma

For $F \in \mathrm{U}(\mathfrak{g})$, write

$$\nabla(F) = \Delta_0(F) - F \otimes 1 - 1 \otimes F.$$

Relative primitive elements

If $\nabla(F) \in \mathrm{U}(\mathfrak{k}) \otimes \mathrm{U}(\mathfrak{g})$, then

$$F \in \mathrm{U}(\mathfrak{k}) + \mathfrak{p}.$$

Applied to the difference of two degree-one lifts, the lemma gives

$$b(y) = \mathcal{J}(y) - \frac{\hbar}{4}[y, C_{\mathfrak{k}}] + \hbar z(y), \qquad z \in \mathrm{End}_{\mathfrak{k}}(\mathfrak{p}),$$

after removing a correction in $\hbar\mathrm{U}(\mathfrak{k})$.

The parameters

The isotropy representation satisfies

$$\mathrm{End}_{\mathfrak{k}}(\mathfrak{p}) = \mathbb{C}\,\mathrm{id}_{\mathfrak{p}} \oplus \mathrm{ad}\,\mathfrak{z}(\mathfrak{k})|_{\mathfrak{p}}.$$

Therefore $z(y) = ay + [K, y]$. If $\mathfrak{z}(\mathfrak{k}) = 0$, only a remains.

case	$\mathfrak{z}(\mathfrak{k})$	$\mathfrak{p}$	$\mathrm{End}_{\mathfrak{k}}(\mathfrak{p})$
non-Hermitian	0	irreducible	$\mathbb{C}\,\mathrm{id}$
Hermitian	$\mathbb{C}z$	$\mathfrak{p}^+ \oplus \mathfrak{p}^-$, weights $\pm\lambda$	$\mathbb{C}\,\mathrm{id} \oplus \mathbb{C}\ \mathrm{ad}(z)$

Table: Isotropy modules for nontrivial symmetric pairs. [Helgason78]

Spectral and central shifts in $\mathfrak{sl}_2$

For $\mathfrak{k} = \mathbb{C}h$ and $\mathfrak{p} = \mathbb{C}e \oplus \mathbb{C}f$, write $K = \lambda h$. Then $[K, e] = 2\lambda e$ and $[K, f] = -2\lambda f$, so

$$b_{a,\lambda h}(e) - b_{0,0}(e) = \hbar(a + 2\lambda)e,$$
$$b_{a,\lambda h}(f) - b_{0,0}(f) = \hbar(a - 2\lambda)f.$$

The meaning of the parameters

The spectral parameter shifts the two coefficients equally. The central parameter shifts them in opposite directions.

They describe distinct embedded realizations of one graded algebra.

Existence and PBW

PBW theorem for every (a, K)

The candidate $B_{a,K}$ satisfies

$$B_{a,K} \cap \hbar \mathrm{Y}_\hbar(\mathfrak{g}) = \hbar B_{a,K}, \qquad B_{a,K}/\hbar B_{a,K} \cong \mathrm{U}(\mathfrak{g}[u]^{\check{\omega}}).$$

Ordered monomials in homogeneous lifts of a basis of $\mathfrak{g}[u]^{\check{\omega}}$ form a $\mathbb{C}[\hbar]$-basis of $B_{a,K}$.

- At $\hbar = 1$, compare word degree with current degree.
- A first discrepancy would give a primitive remainder of a forbidden classical Serre weight.

Uniqueness of the graded algebra

Spectral translation

The Yangian automorphism

$$x \longmapsto x, \qquad J(x) \longmapsto J(x) + c\hbar x$$

sends $B_{a,K}$ to $B_{a+c,K}$.

Central translation: $D = K_1 - K_0$

A graded algebra isomorphism $B_{a,K_0} \longrightarrow B_{a,K_1}$ is

$$x \longmapsto x + (D, x)1, \qquad b_{a,K_0}(y) \longmapsto b_{a,K_1}(y).$$

Its inverse uses $-D$.

The central translation changes the classical generators in $\mathfrak{z}(\mathfrak{k})$.

The reduced Cartan relation

Take linear generating maps $\iota : \mathfrak{k} \to B_0$, $b : \mathfrak{p} \to B_1$, with

$$[\iota(x), \iota(x')] = \iota([x, x']), \qquad [\iota(x), b(y)] = b([x, y]).$$

Let $q_i = \frac{1}{2}\theta_i^2 + \frac{1}{4}\sum_{\alpha\in\Phi^+}(\alpha, \alpha_i)\{b_\alpha, b_{\tau\alpha}\}$.

Presentation theorem: split and quasi-split pairs

At the reference central parameter $K_\circ$, the additional relations are

$$[b(t_i), b(t_j)] = \hbar^2\iota([q_j, q_i]), \qquad t_i \in \mathfrak{h} \cap \mathfrak{p},$$

where $q_i \in \mathrm{U}(\mathfrak{k})$ are explicit quadratic polynomials.

Split types exclude A_1. Nontrivial quasi-split types A, D, E_6 exclude A_2.

It is equivalent to the minimalistic presentation in [L26,Huang-L26]

How to find the relations

In the constructed coideal, consider the defect

$$R_{ij} = [b(t_i), b(t_j)] - \hbar^2 [q_j, q_i].$$

- Expand the commutator using the normalized embedding in $\mathrm{Y}_\hbar(\mathfrak{g})$.
- The Yangian Cartan relation cancels its weight-zero part.
- The remaining terms have strictly positive weights. Saturation and classical PBW force $R_{ij} = 0$.

The proof uses the Yangian relations and the classical twisted current algebra.

The conclusions

- GWX prove uniqueness of the homogeneous Hopf quantization of $\mathfrak{g}[u]$.
- For symmetric pairs, we prove existence and uniqueness of homogeneous coideal quantizations in a fixed Yangian.
- The embeddings carry spectral and central parameters. All give isomorphic graded algebras.

Thank you!